

Product Overview

Protpure SP Agarose resins are strong cation exchange chromatography media based on a 6% highly cross-linked agarose matrix. Available in Fast Flow, Precise and High Resolution grades, they provide a range of performance options from high-throughput capture and intermediate purification to high-resolution polishing applications.

They are designed for high-resolution purification of proteins and other biomolecules, offering high binding capacity, excellent selectivity, robust chemical stability, reproducibility, and scalability for bioprocess development and manufacturing.



High Binding Capacity
Consistent & Reproducible



Robust Chemical Stability
Wide pH & CIP Range



Scalable Performance
From Lab to Manufacturing



Excellent Flow Properties
For High-Resolution Separations



Agarose-Based
Bioprocess Platform

Comparative Performance Overview

Parameter	SP Agarose Fast Flow	SP Agarose Precise	SP Agarose HR	Typical Market Specification
BioProcess Resin	Yes	Yes	Yes	Yes
Ion Exchanger Type	Strong cation exchanger	Strong cation exchanger	Strong cation exchanger	Strong cation exchanger
Ionic Capacity (IEC)	0.18–0.25 mmol (H ⁺)/1 mL settled resin	0.18–0.25 mmol (H ⁺)/1 mL settled resin	0.18–0.25 mmol (H ⁺)/1 mL settled resin	0.18–0.25 mmol (H ⁺)/1 mL settled resin
Ligand	Sulfopropyl	Sulfopropyl	Sulfopropyl	Sulfopropyl
Matrix	6% spherical cross-linked agarose	6% spherical cross-linked agarose	6% spherical cross-linked agarose	6% cross-linked agarose, spherical
Particle Size Range	45–165 µm	45–105 µm	25–55 µm	20–50 µm
Particle Size (d50V)	~90 µm	~65 µm	~35 µm	30–40 µm
Dynamic Binding Capacity (10% breakthrough)	≈100 mg Lysozyme/mL resin	≈120 mg Lysozyme/mL resin	≈130 mg Lysozyme/mL resin	≥100 mg Lysozyme/mL resin
Pressure/Flow Specification (0.1 MPa, 15 cm bed height)	250–450 cm/h	100–300 cm/h	70–120 cm/h	50–120 cm/h
pH Stability (CIP)	2–14	2–14	2–14	2–14
pH Stability (Operational)	2–12	2–12	2–12	2–12
Chemical Stability	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol
Storage Conditions	4–30°C, 20% Ethanol	4–30°C, 20% Ethanol	4–30°C, 20% Ethanol	4–30°C, 20% Ethanol

Applications (SP Agarose Resins)



- Monoclonal antibody purification
- Recombinant protein purification
- Enzyme purification
- Vaccine and viral vector downstream processing
- Host cell protein (HCP) removal
- Intermediate purification and polishing
- Purification of positively charged biomolecules
- Process development and manufacturing-scale chromatography

Key Benefits (SP Agarose Resins)

- ✓ High binding capacity and excellent selectivity
- ✓ High-resolution separations with superior purity
- ✓ Excellent flow properties for process-scale chromatography
- ✓ Robust chemical and alkaline stability
- ✓ Consistent lot-to-lot performance
- ✓ Suitable for scale-up from R&D to commercial manufacturing
- ✓ Agarose-based platform for reliable biomolecule purification



Evaluation samples and application support available upon request.
Contact us to discuss your process requirements.

